Name: _____ Reg #: _____ Section: _____

National University of Computer and Emerging Sciences, Lahore Campus

Course:	**Natural Language Processing**	**Course Code:**	**CS 535**
Program:	**MS (Data Science)**	**Semester:**	**Fall 2021**
Duration:	**80 mins**	**Total Marks:**	**38**
Paper Date:	**10-Nov-21**	**Weight**	**20%**
Section:	**DS**	**Page(s):**	**6**
Exam:	**Midterm Exam**		

Instruction/Notes: Attempt the examination on the question paper and write concise answers. You can use extra sheet for rough work. Do not attach extra sheets used for rough with the question paper. Don't fill the table titled Questions/Marks.

Questions	1	2	3	4	5	6	7	Total
Marks	/8	/5	/4	/5	/3	/6	/7	/38

Q1) Answer the following multiple choice questions. Suppose you have the following training data for Näıve Bayes. Encircle correct option. [8 Marks]

The movie was terrible [LABEL= -]
The movie could be more interesting if duration was shorter [LABEL= -]
It was boring in the beginning but fun overall. [LABEL= +]

A) What is the unsmoothed maximum likelihood estimate of P(-)for this data?
i. 1/3 ii. 1/2 iii. 2/3 iv. 1

(B) What is the unsmoothed maximum likelihood estimate of P(fun |+)for this data? $\frac{\text{count}(\text{fun})}{n(+)} = \frac{1}{9}$
i. 1/23 ii. 1/9 iii. 1/18 iv. 1/3

(C) Suppose we are given an unseen input sentence "the movie". What is the joint probability P(−,the movie)? $= \left(\frac{2}{3}\right)\left(\frac{2}{18}\right)$
i. 2/18 ii. 1/147 iii. 2/147 iv. 4/23

(D) What prediction will the model make on sentence "the movie"? $P(-|\text{the movie}) = \frac{2}{18}$ ✓ $P(+|\text{the movie}) = \frac{0}{8} = 0$
i. Positive ii. Negative

Q2) Calculate the TFIDF for the terms listed below for documents 1. There are 100 documents in a collection. The number of times each of these terms occur in documents 1 to 3 as well as the number of documents in the collections are listed below. Use this information to fill in the TFIDF scores for Doc 1 in the table below. [5 Marks]

Number of Documents Containing Terms:
_ Table: 30
_ Bottle: 10
_ Pen: 50

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	Raw Term Counts		
	Doc 1	Doc 2	Doc 3
Table	4	54	3
Bottle	7	5	30
Pen	25	34	9

Fill in the table below and show all working.

	Tf.IDF for terms in Doc 1		
Table	$1+\log 4$		
Bottle	7		
Pen	25		
	Raw TF	IDF	TF×IDF

$\log\left(\frac{100}{30}\right)$

$\log\left(\frac{100}{10}\right)$

$\log\left(\frac{100}{50}\right)$

Q3) TRUE / FALSE ? [4 Marks]

A. Recurrent neural networks can handle a sequence of arbitrary length,while feedforward neural networks can not.
B. Training recurrent neural networks is hard because of vanishing and exploding gradient problems.
C. Dropout is an effective way of solving exploding gradient problem.
D. Gated recurrent units (GRUs) have more parameters than LSTMs.

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Q4) Suppose following table gives word vectors for the words of ”Table”, Bottle”, and ”Pen”. Each vector has 4 dimensions and weights in each dimeniosn are given in table. Calculate cosine similarity between vectors of ”Table” and ”Bottle”. [5 Marks]

	Apple	Sugar	Chair	Key
Table	4	5	1	6
Bottle	3	3	3	10
Pen	1	11	12	1

cosine (bottle, pen) = $\frac{3+33+36+10}{\sqrt{3^2+3^2+3^2+10^2} \cdot \sqrt{1^2+11^2+12^2+1^2}} = 0.45$

Q5) In statistical language modeling, you have read about the concept of smoothing. Which type of documents need more smoothing, long or short? Why? [3 Marks]

Short, bcoz it has less n-gram likely than long

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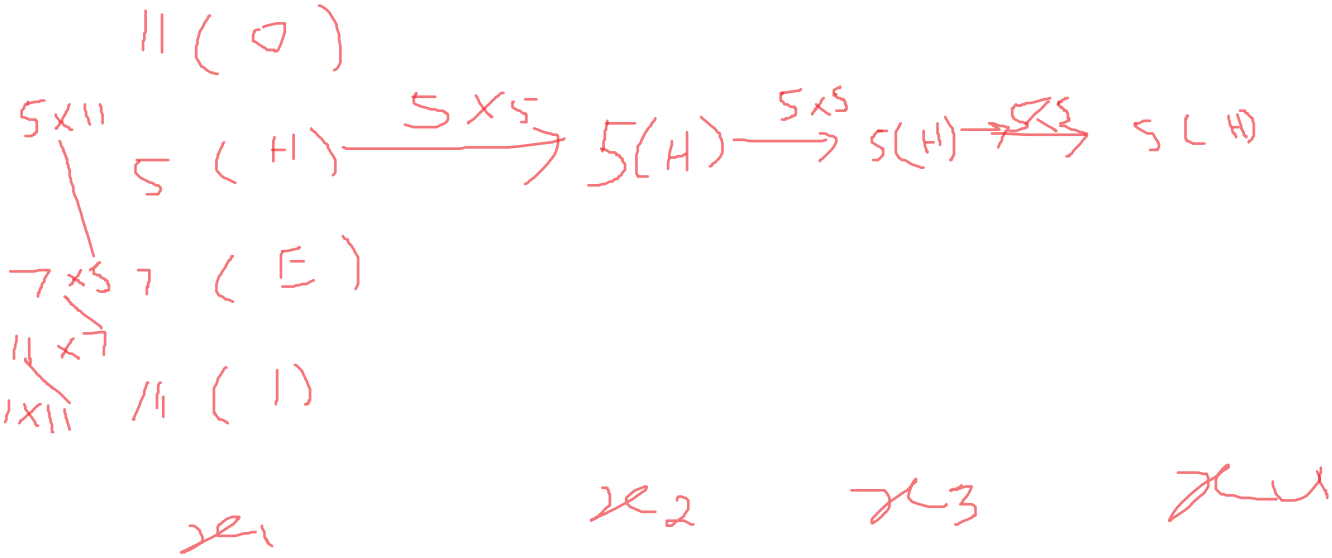
Q6) (a) Suppose we have following language model: [4+2 = 6 Marks]

• input sequence of length 4 (lets say 4 words).
• Hidden layer units are 5.
Embedding vector size = 7
• V = vocabulary = 11

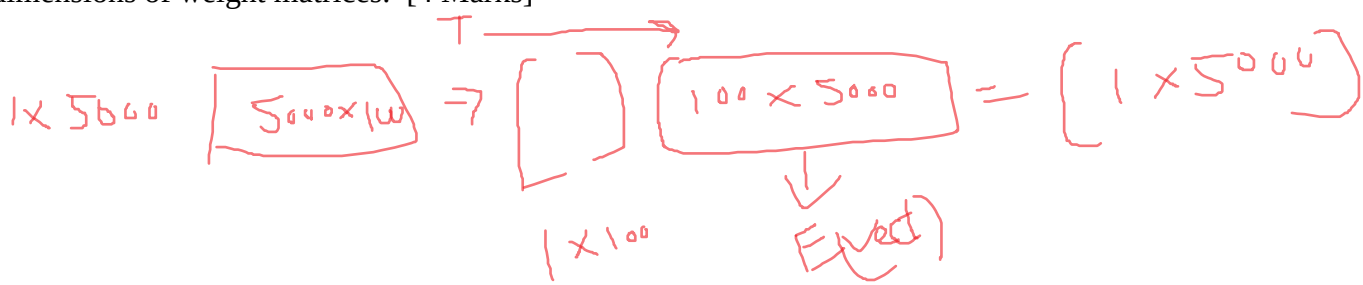
i. Draw RNN architecture diagram with dimensions of all layers and weight matrices


11 (O)
5 (H)
5 (H)
5 (H

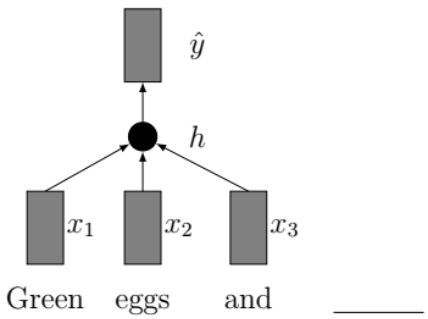
ii. Give the update equations for a simple RNN unit in terms of x, y, and h (input x, output y, and recurrent state h). Assume it uses tanh non-linearity.

Q7) (a) Suppose we have a collection with following statistics:
Total words = 100,000
Vocabulory = 5000

We want to train a WordToVec model on this collection. The dimension of each word vector should be 100. Draw WordToVec architecture diagram with dimensions of input, hidden, and output layers, and dimensions of weight matrices. [4 Marks]



Q7) (b) A feedforward neural network language model (NNLM) can be used as another architecture for training word vectors. This model tries to predict a word given the N words that precede it. To do so, we concatenate the word vectors of N previous words and send them through a single hidden layer of size H with a tanh nonlinearity and use a softmax layer to make a prediction of the current word. The size of the vocabulary is V. The model is trained using a cross entropy loss for the current word. Let the word vectors of the N previous words be x1,x2,...,xN, each a column vector of dimension D, and let y be the one-hot vector for the current word.



$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_N \end{bmatrix}$$

$$\mathbf{h} = \tanh(W\mathbf{x} + \mathbf{b})$$

$$\hat{\mathbf{y}} = \text{softmax}(U\mathbf{h} + \mathbf{d})$$

$$J = \text{CE}(\mathbf{y}, \hat{\mathbf{y}})$$

$$CE = -\sum_i y_i \log(\hat{y}_i).$$

State two important differences between NNLM the WordToVec language model we learned in class. Explain how each might affect the word vectors learned. [3 Marks]

word2vec word vectors are obtained by training the model on previous and next context words (context window). NNLM word vectors will only have context from previous context words.